

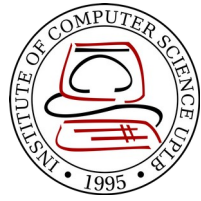
CMSC 137

Data Communications and Networking

Lecture Module
08 – Analog Transmission



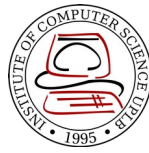
University of the Philippines
LOS BAÑOS



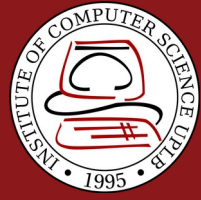
Module 8: Analog Transmission

Rozano S. Maniaol
Lecturer

Introduction



- If digital transmission is desirable, why do we need analog transmission?
 - If available channel bandwidth is scarce, hence we are forced to use a **bandpass channel**
- Traditionally,
 - Converting digital data to a bandpass analog signal is called **digital to analog conversion**
 - Converting a low-pass analog signal to a bandpass analog signal is called **analog to analog conversion**

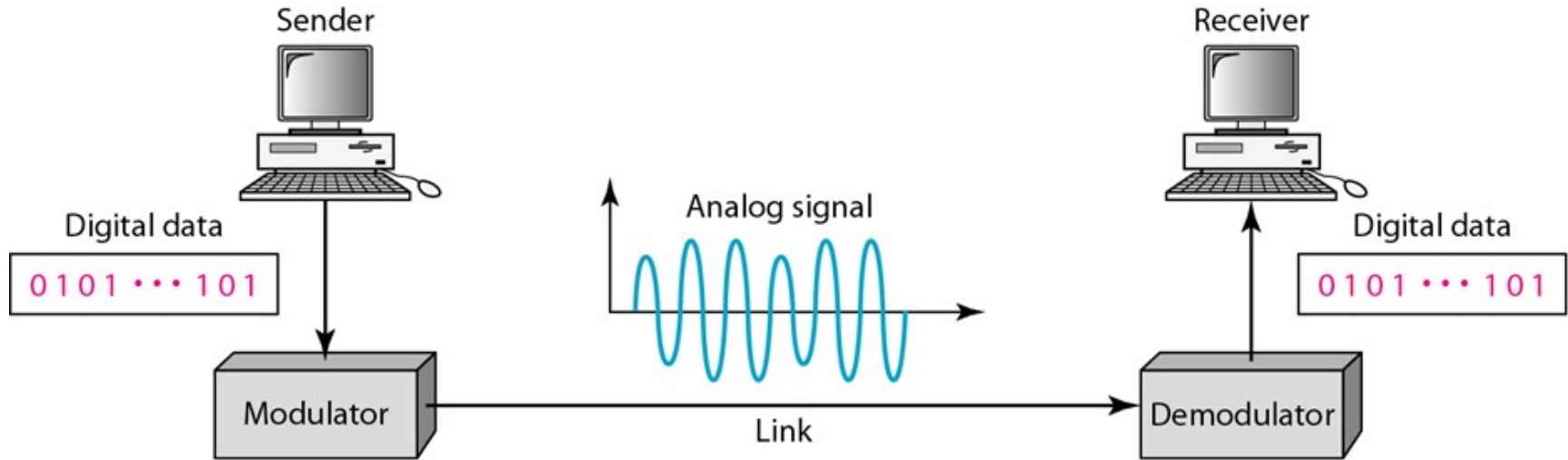


Digital to Analog Conversion

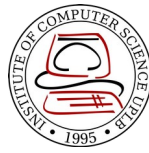
Digital to Analog Conversion



- Changing the characteristics of an analog signal based on the information in digital data

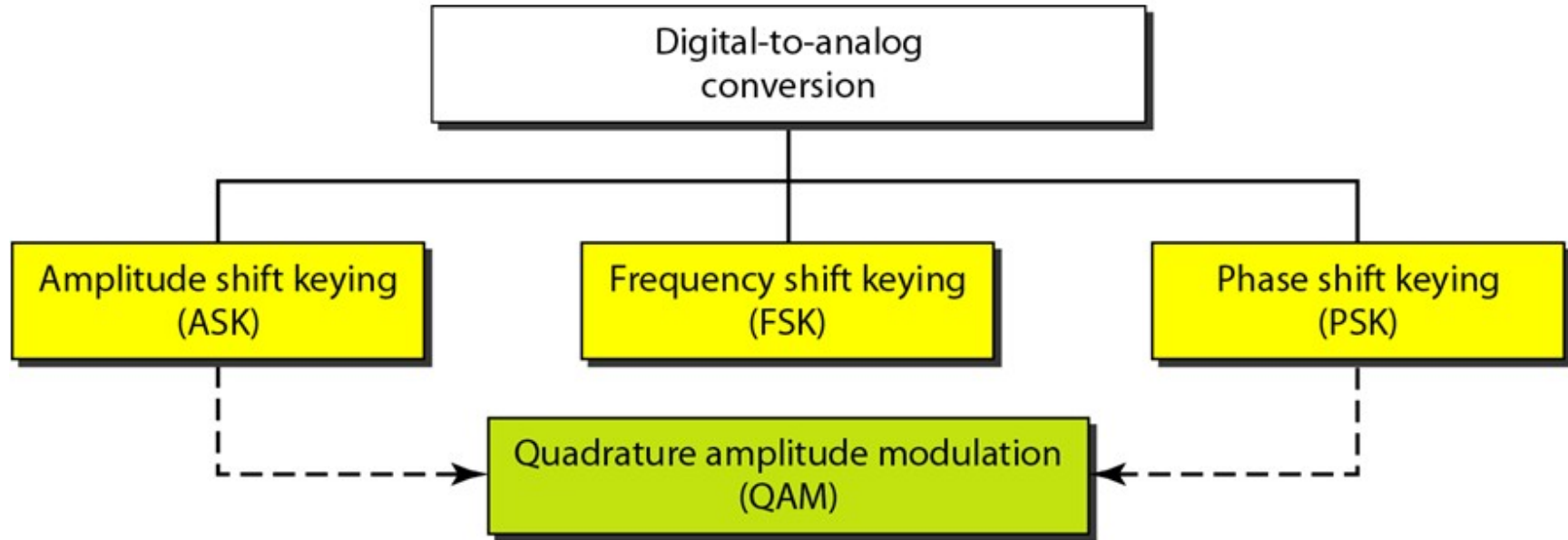


Definitions

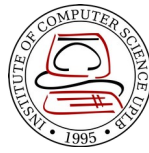


- **Bit rate** is the number of bits per second. **Baud rate** is the number of signal elements per second. In analog transmission of digital data, the baud rate is less than or equal to the bit rate
- **Carrier signal** (or carrier frequency) f_c
 - High-frequency signal that acts as a base for the information signal that is produced by the sending device in analog transmission
- **Modulation** (or shift keying) – modifying the characteristics of a carrier frequency to send data

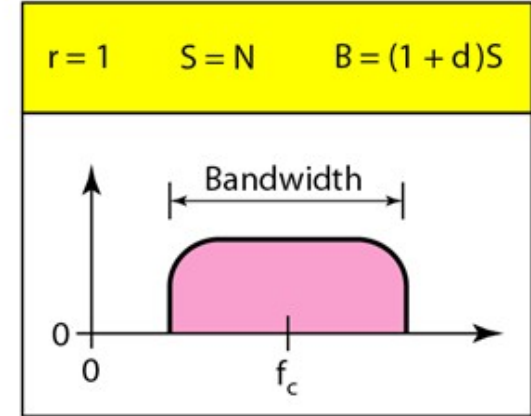
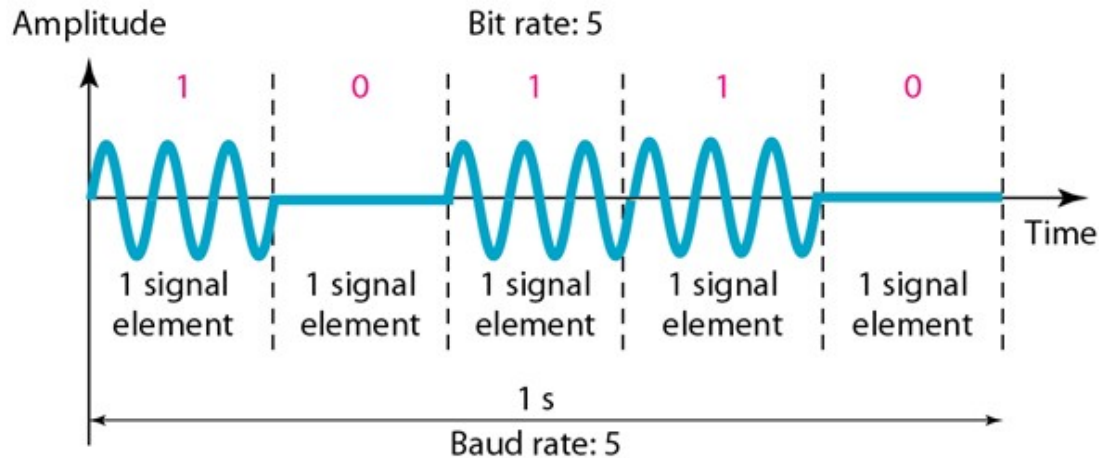
Types of Digital to Analog Conversion



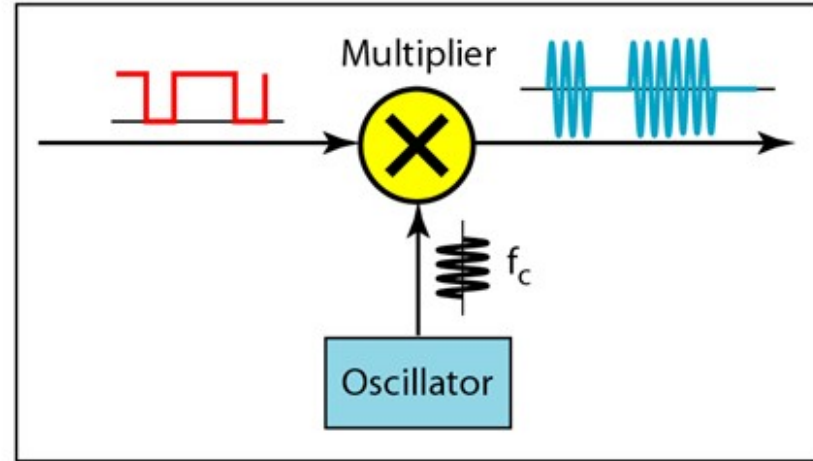
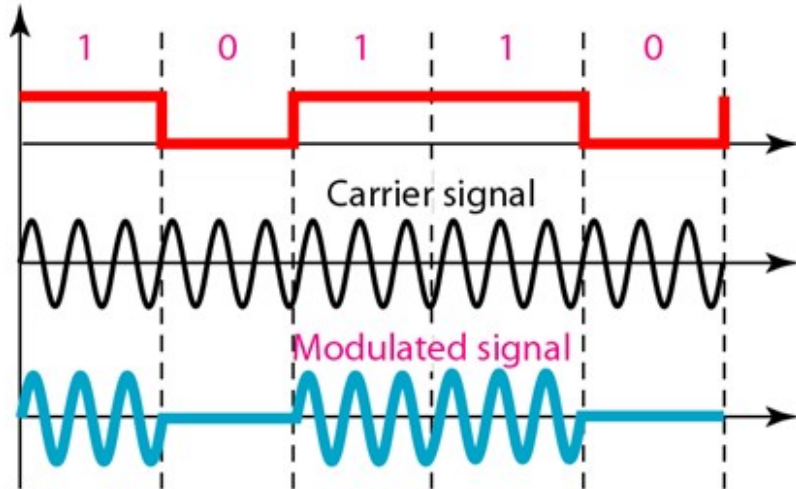
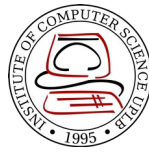
Amplitude Shift Keying (ASK)



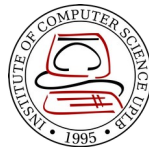
- Amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while amplitude changes



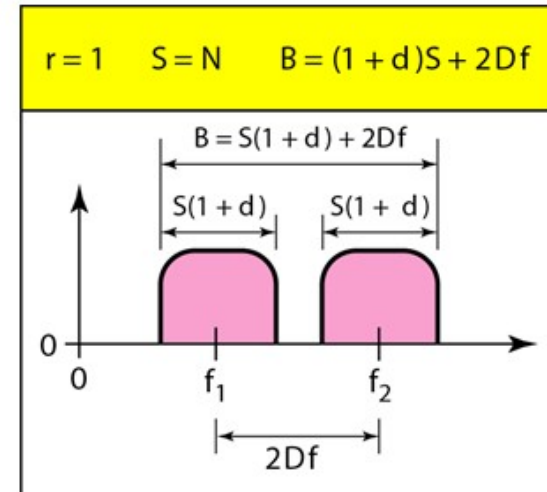
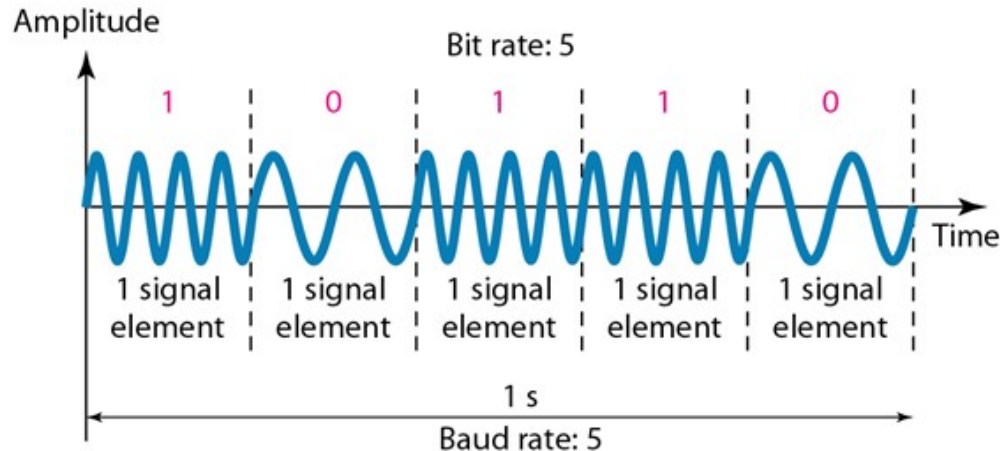
Implementation of BASK



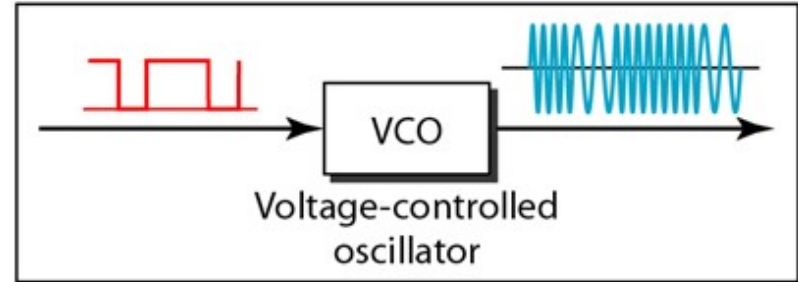
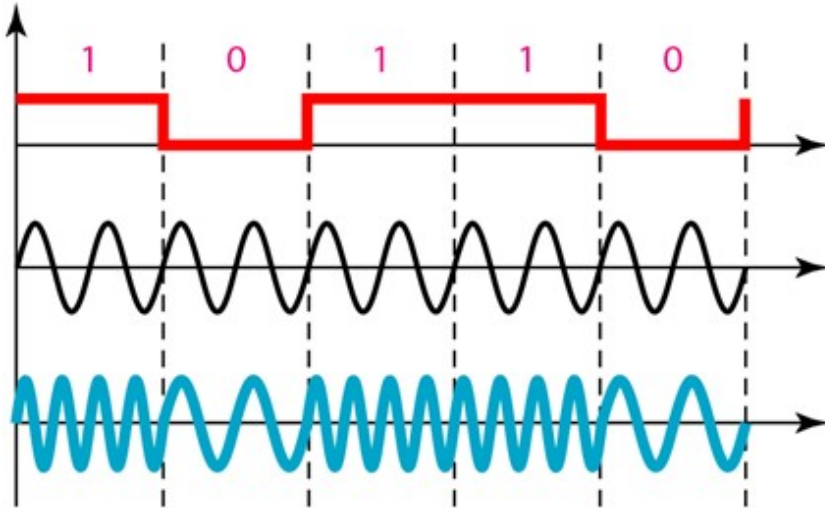
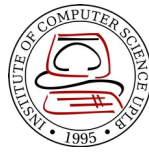
Frequency Shift Keying (FSK)



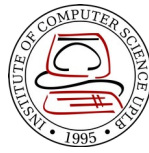
- Frequency of the carrier signal is varied to represent data
- Both peak amplitude and phase remain constant for all signal elements



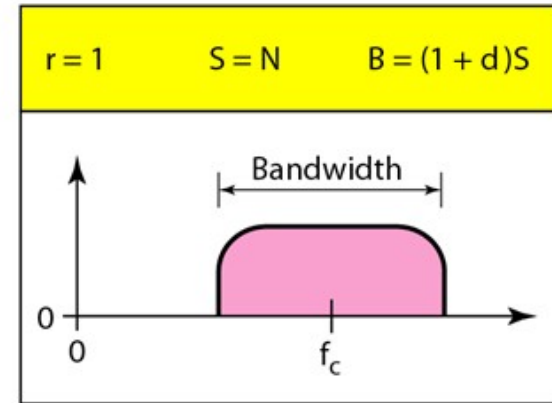
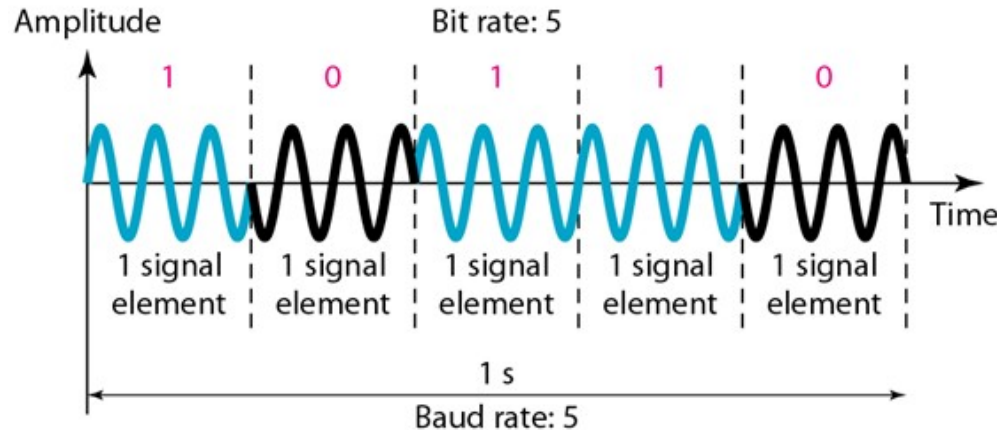
Implementation of BFSK



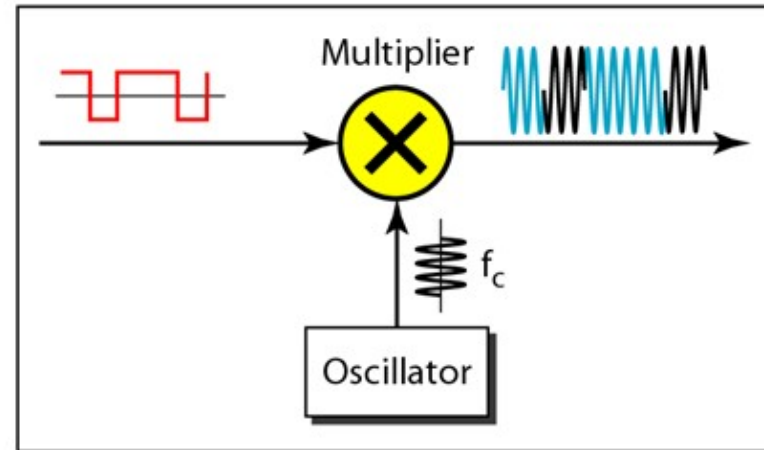
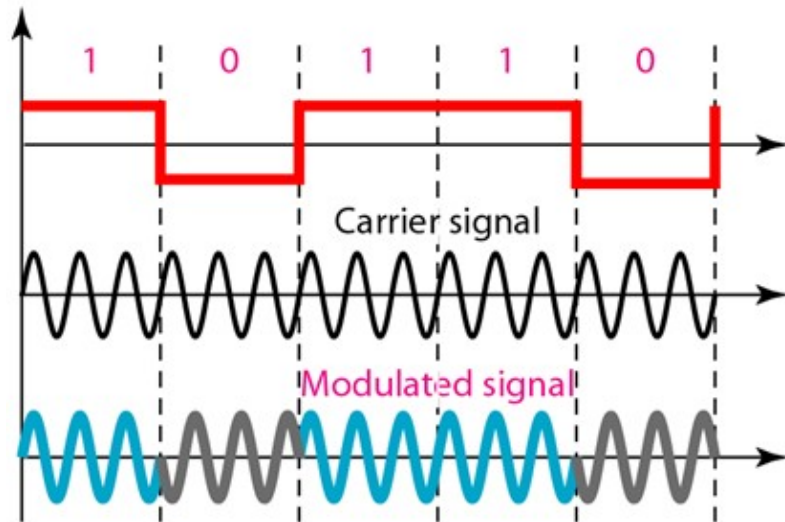
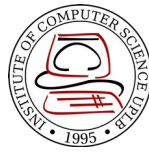
Phase Shift Keying (PSK)



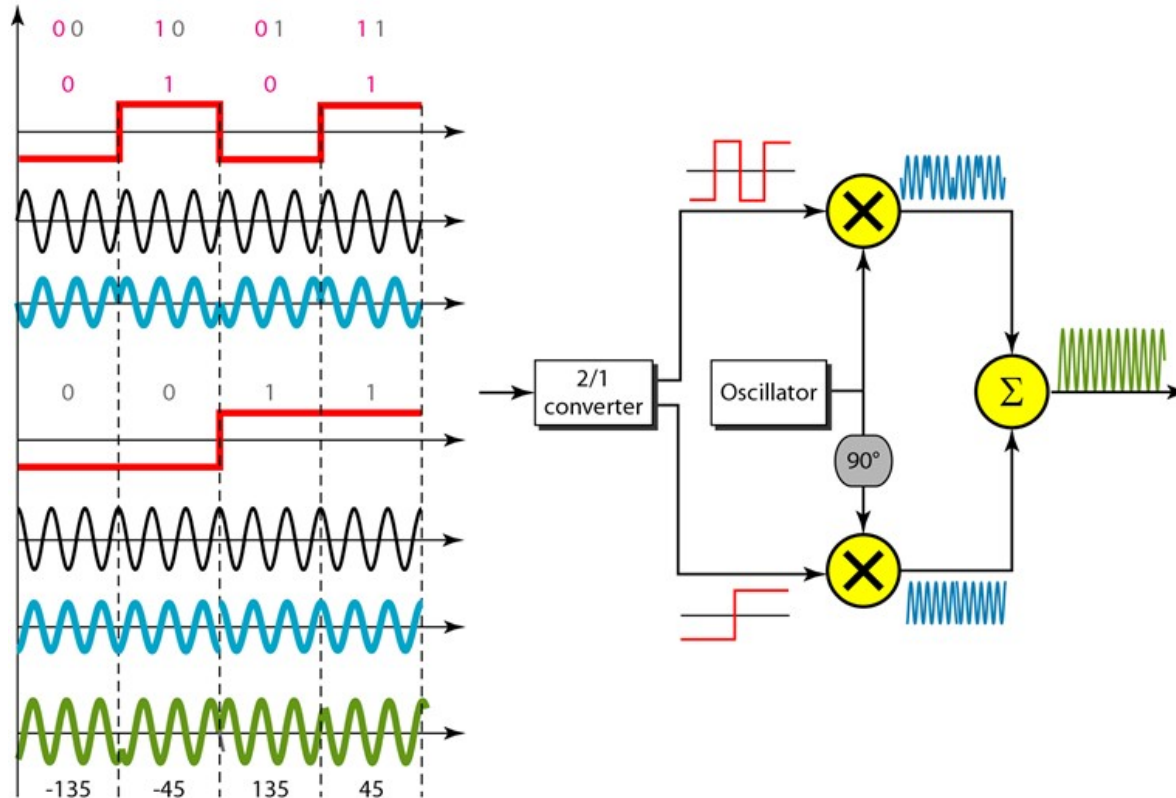
- The phase of the carrier is varied to represent two or more different signal elements
- Both peak amplitude and frequency remain constant



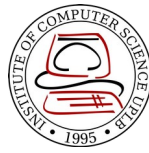
Implementation of BPSK



Implementation of QPSK

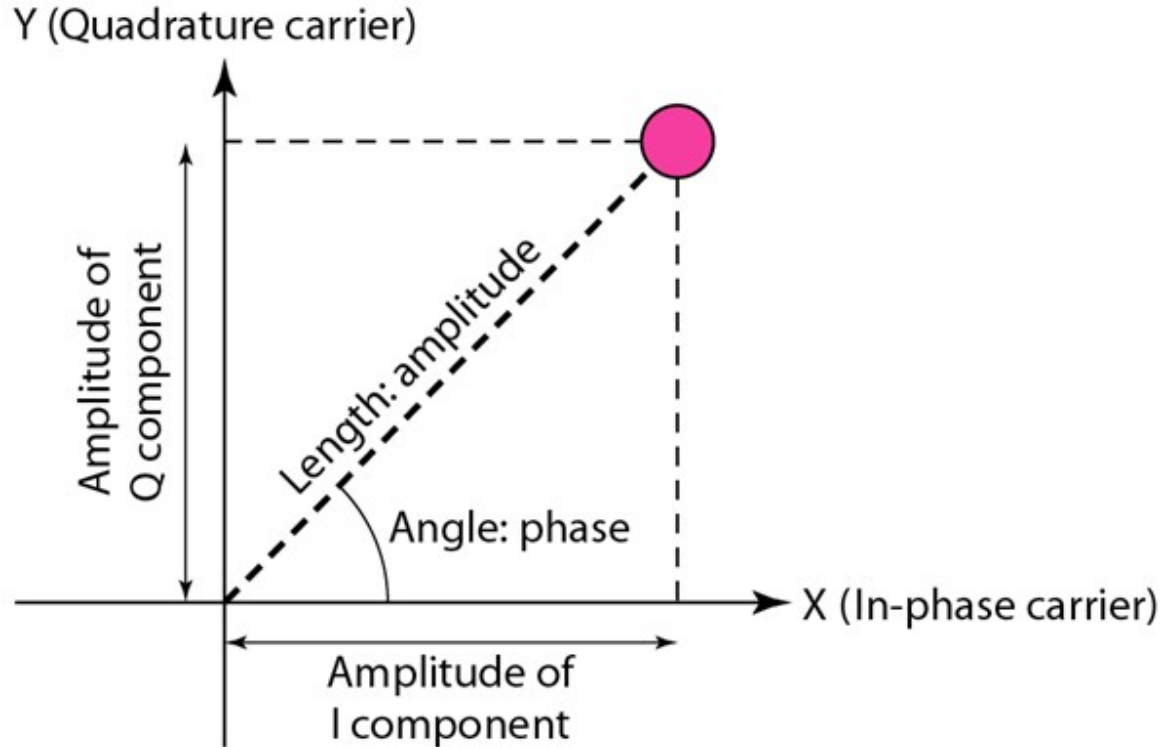


Constellation Diagram

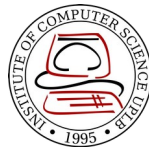


- Helps to define the amplitude and phase of a signal element when we are using two carriers, one in quadrature of the other
- Useful for dealing with multilevel ASK, PSK or QAM
- The x-axis represents the **in-phase carrier** and the y-axis represents the **quadrature carrier**
- A signal element is represented by a dot
- The bit or combination of bits it can carry is often written next to it

Constellation Diagram

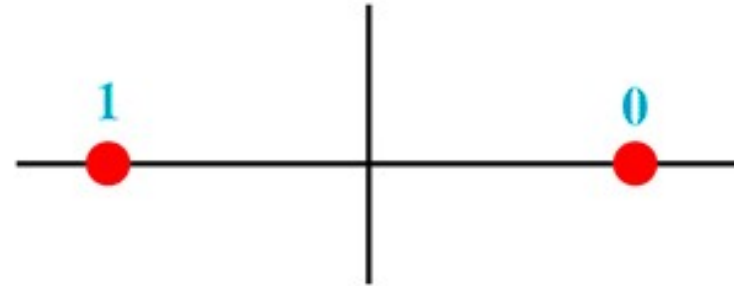


PSK Constellation Diagram



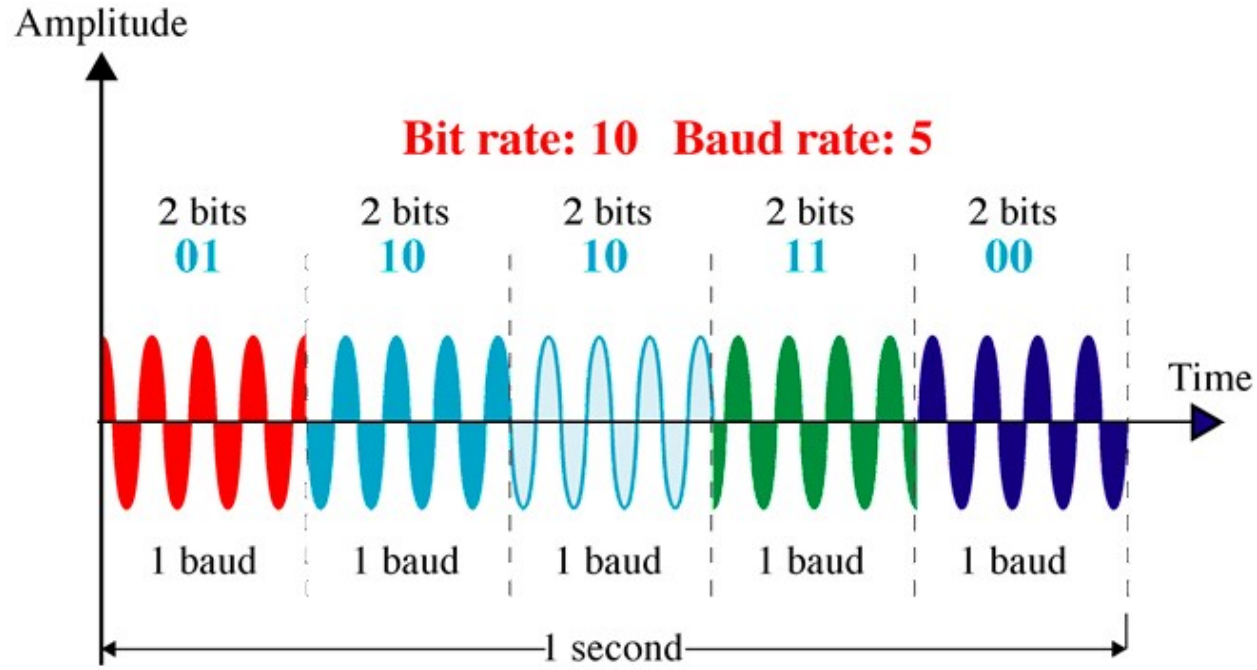
Bit	Phase
0	0
1	180

Bits

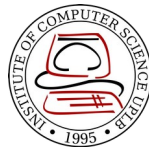


Constellation diagram

4-PSK

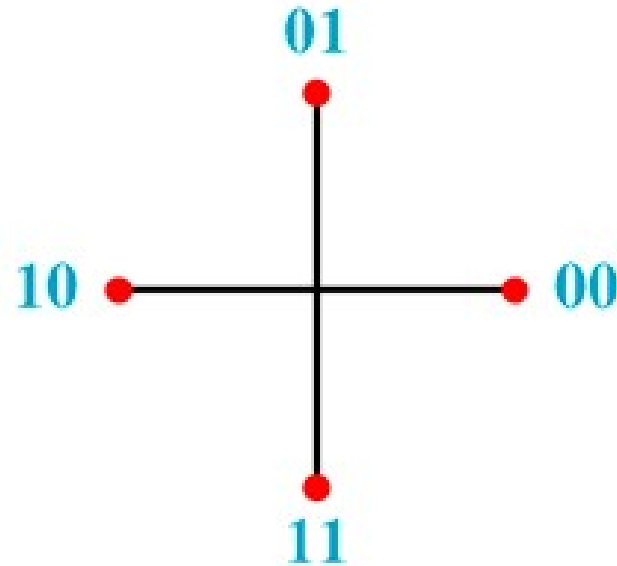


4-PSK Constellation Diagram



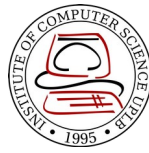
Dibit	Phase
00	0
01	90
10	180
11	270

Dibit
(2 bits)



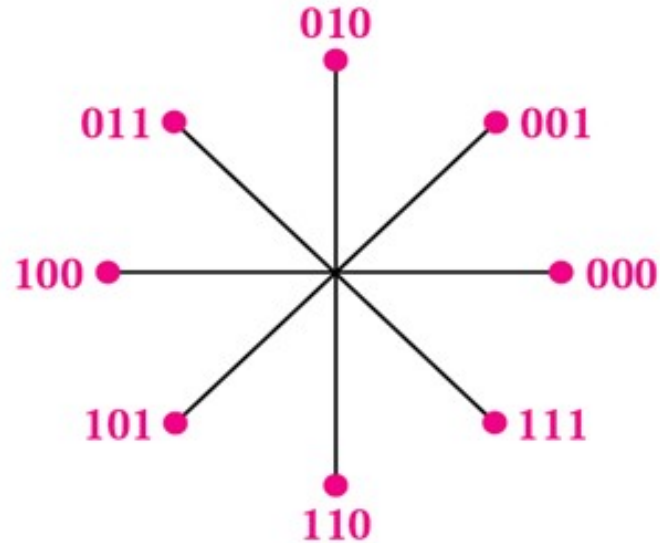
Constellation diagram

8-PSK Constellation Diagram



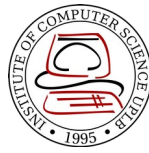
Tribit	Phase
000	0
001	45
010	90
011	135
100	180
101	225
110	270
111	315

Tribits
(3 bits)

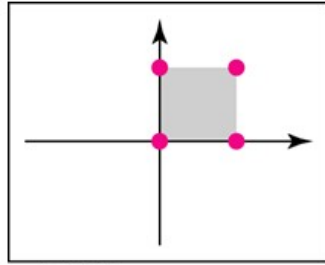


Constellation diagram

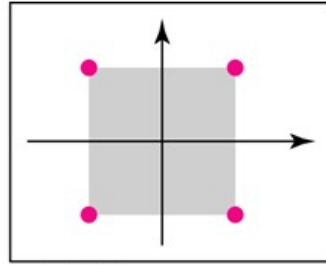
Quadrature Amplitude Modulation (QAM)



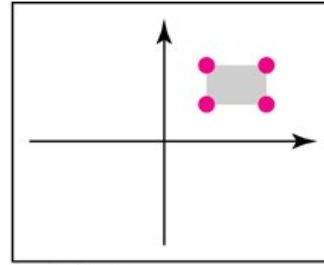
Combination of ASK and PSK



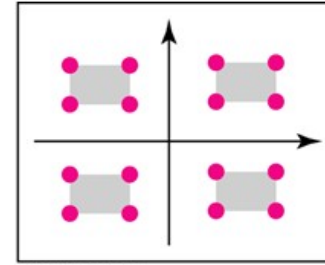
a. 4-QAM



b. 4-QAM

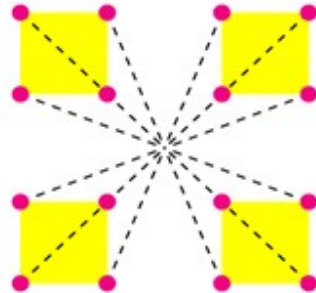


c. 4-QAM



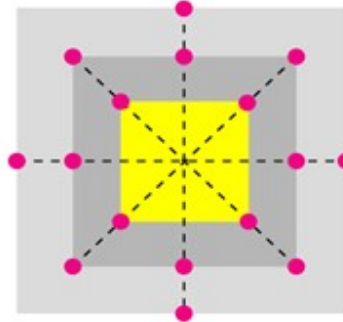
d. 16-QAM

3 amplitudes, 12 phases



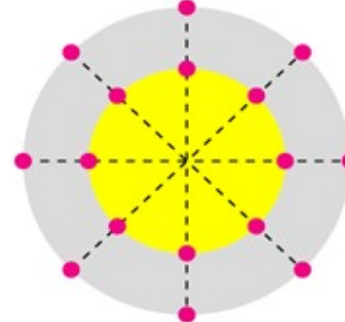
16-QAM

4 amplitudes, 8 phases



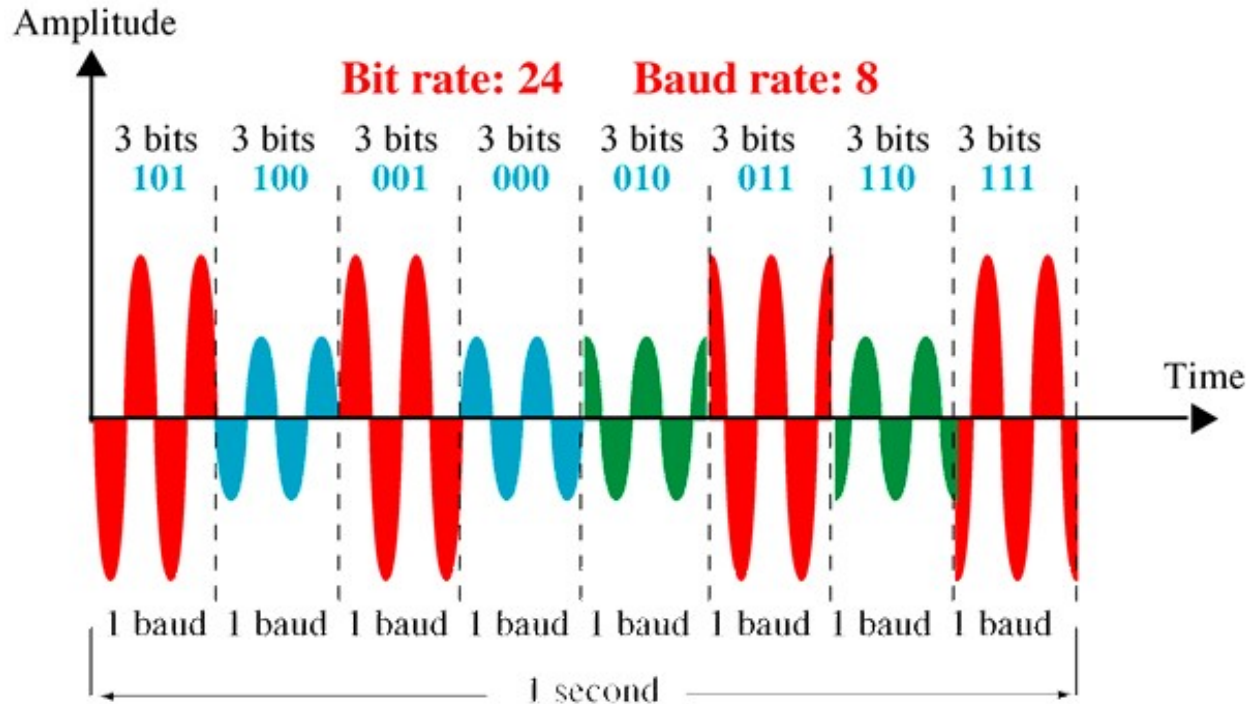
16-QAM

2 amplitudes, 8 phases

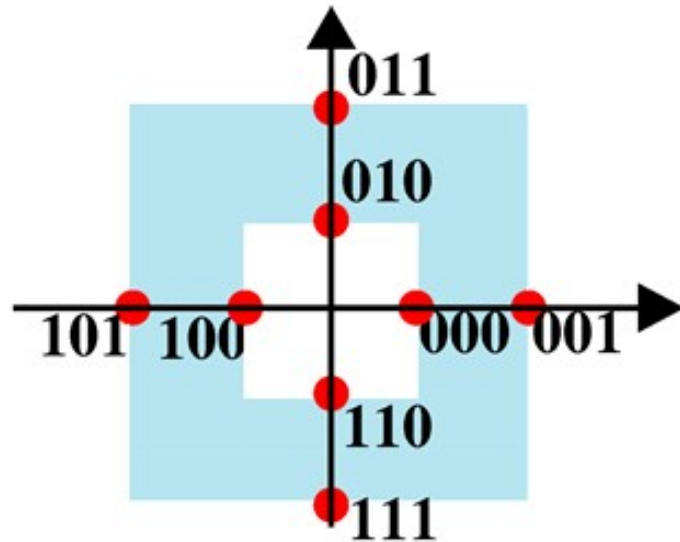


16-QAM

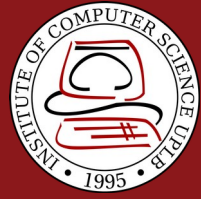
8-QAM Signal



8-QAM Constellation Diagram

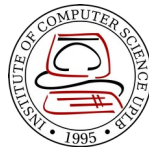


8-QAM
2 amplitudes, 4 phases

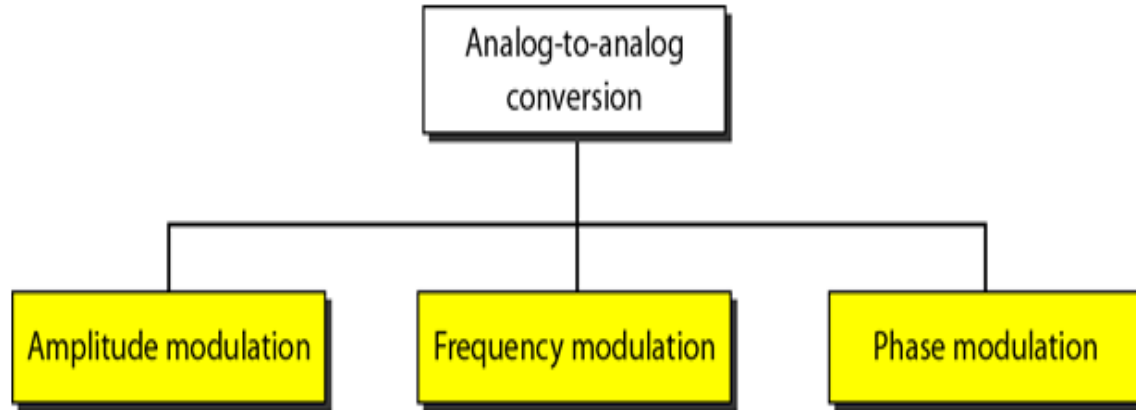


Analog to Analog Conversion

Analog to Analog Conversion



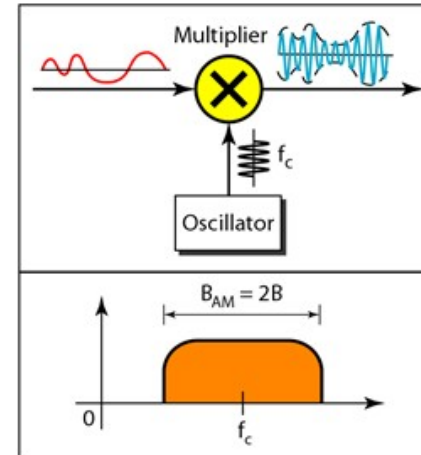
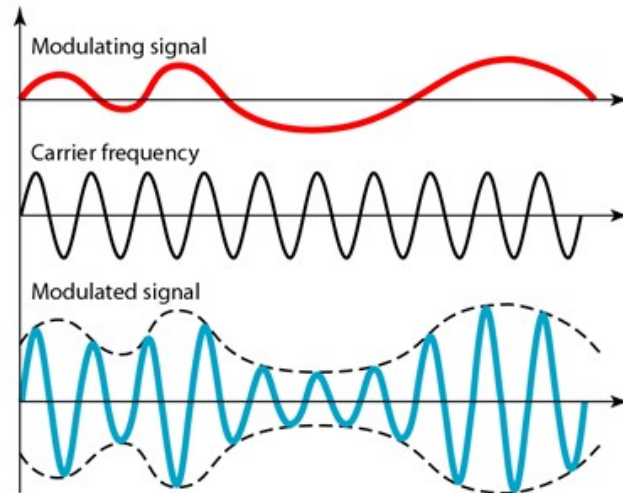
- Representation of analog information by an analog signal
- Modulation is needed if medium is bandpass in nature or if only a bandpass channel is available



Amplitude Modulation (AM)



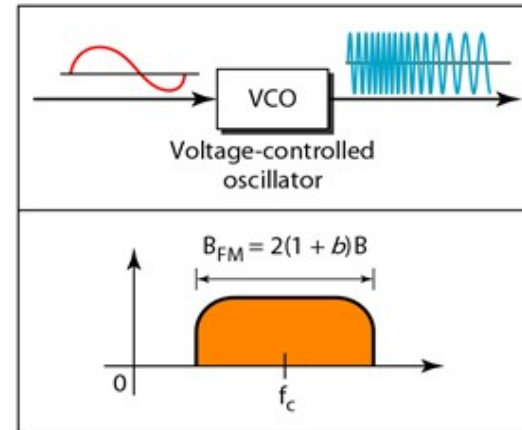
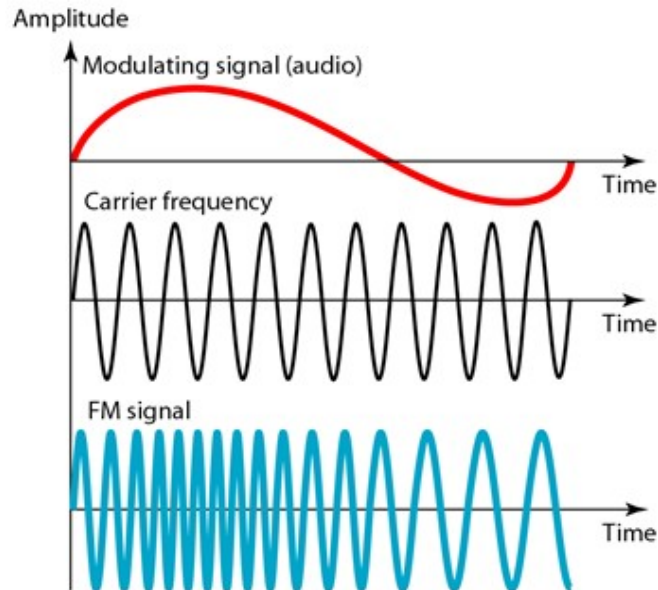
- Carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal
- The modulating signal is the envelope of the carrier



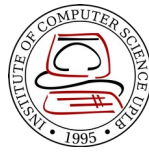
Frequency Modulation (FM)



- Frequency of the carrier signal is modulated to follow the changing voltage level (amplitude) of the modulating signal



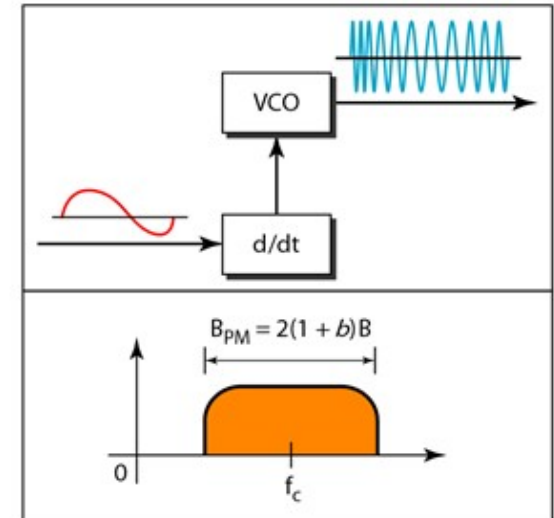
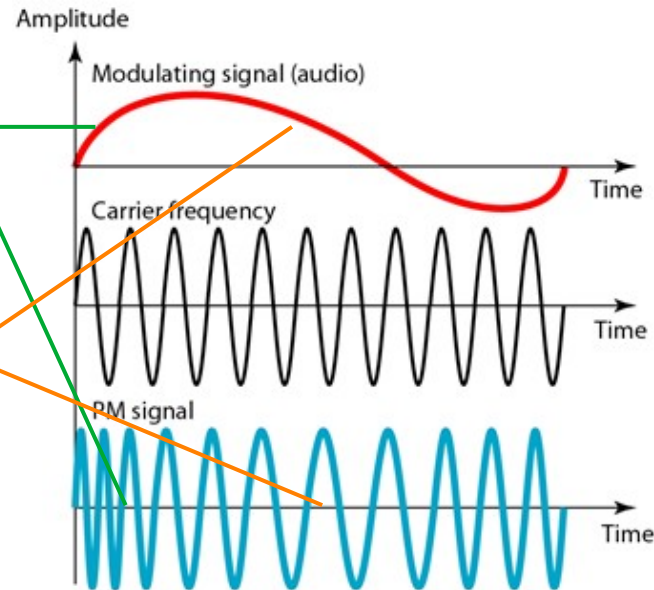
Phase Modulation (PM)



- Phase of the carrier signal is modulated to follow the changing voltage level of the modulating signal

As the amplitude increases,
the magnitude of the phase
lead increases

As the amplitude decreases,
the magnitude of the phase
lag increases



FM vs PM



- In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal.
- In PM, the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal.
- PM is used in mobile communication systems, while FM is used mainly in FM broadcasting

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.